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# New York Government Spending Your Dime

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# Question:

Does government spending within NY towns come more from local taxes or government aid? (as seen through 3 metrics)

Specifically measuring:

- General Government Spending
- Public Safety
- Social Services

# Data

Data Source: Office of the New York State Comptroller, Local Government Bulk Data

- 9941 observations across 914 NY Towns
- 10 years worth of data (2013–2023)

Converted to panel by appending each year into one dataset

# gen\_gov

Total amount of expenditures for services provided by the governmental entity for the benefit of the public or governmental body as a whole.

Includes:

- Administration
- Zoning and planning
- Operations
- Judgments
- County distribution of sales tax
- Miscellaneous general government

# public\_safety

Total amount of expenditures for police, fire and other public safety services.

Includes:

- Public safety administration
- Police
- Fire protection
- Emergency response
- Correctional services
- Disaster response
- Homeland security and civil defense
- Miscellaneous public safety

# social\_services

Total amount of expenditures for providing public assistance for individuals/families economically unable to provide essential needs for themselves.

Includes:

- Social service administration
- Financial assistance
- Medicaid
- Non-Medicaid medical assistance
- Housing assistance, employment services
- Youth services
- Public facilities
- Miscellaneous social services

# fed\_aid

Total amount of revenues derived from the federal government

Includes:

- Assistance with education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous federal Aid

# state\_aid

Total amount of revenues derived from State Aid.

Includes:

- Education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous State Aid



# local\_taxes

Total amount of tax revenue within town.

Includes:

- Property Tax
- Non-Property Tax (city income tax)
- Other property Tax (gains from sale of tax-acquired property)
- Sales & Use Tax (Sales Tax)

# Clustering Standard Errors

Looks at each individual entity's standard error instead of lumping them all together.

- Not clustering underestimates standard errors

Coefficients don't change when clustering, only standard errors

# Clustering Standard Errors

With Clustered SE:

| gen_gov       | Coefficient | Robust<br>std. err.               | t     | P> t  | [95% conf. interval] |          |
|---------------|-------------|-----------------------------------|-------|-------|----------------------|----------|
| fed_aid       | 20.54548    | 3.27292                           | 6.28  | 0.000 | 14.12142             | 26.96955 |
| state_aid     | 23.39038    | 8.584306                          | 2.72  | 0.007 | 6.541171             | 40.23959 |
| local_taxes   | -.956704    | .6695676                          | -1.43 | 0.153 | -2.270926            | .357518  |
| calendar_year |             |                                   |       |       |                      |          |
| 2014          | 2499028     | 2363468                           | 1.06  | 0.291 | -2139968             | 7138024  |
| 2015          | 1848775     | 2340137                           | 0.79  | 0.430 | -2744428             | 6441978  |
| 2016          | 2712811     | 2514260                           | 1.08  | 0.281 | -2222158             | 7647780  |
| 2017          | -3697988    | 3428213                           | -1.08 | 0.281 | -1.04e+07            | 3030881  |
| 2018          | -4038495    | 5619114                           | -0.72 | 0.473 | -1.51e+07            | 6990658  |
| 2019          | -6561880    | 5381499                           | -1.22 | 0.223 | -1.71e+07            | 4000885  |
| 2020          | -7254732    | 7048451                           | -1.03 | 0.304 | -2.11e+07            | 6579912  |
| 2021          | 4151582     | 6145287                           | 0.68  | 0.499 | -7910338             | 1.62e+07 |
| 2022          | -2700332    | 5738988                           | -0.47 | 0.638 | -1.40e+07            | 8564108  |
| 2023          | 2601491     | 7255519                           | 0.36  | 0.720 | -1.16e+07            | 1.68e+07 |
| _cons         | -1.87e+07   | 5542937                           | -3.38 | 0.001 | -2.96e+07            | -7855774 |
| sigma_u       | 52823354    |                                   |       |       |                      |          |
| sigma_e       | 65090374    |                                   |       |       |                      |          |
| rho           | .39708005   | (fraction of variance due to u_i) |       |       |                      |          |

Without Clustered SE:

| gen_gov       | Coefficient | Std. err.                         | t     | P> t  | [95% conf. interval] |           |
|---------------|-------------|-----------------------------------|-------|-------|----------------------|-----------|
| fed_aid       | 20.54548    | .6007752                          | 34.20 | 0.000 | 19.36731             | 21.72366  |
| state_aid     | 23.39038    | 1.409932                          | 16.59 | 0.000 | 20.62538             | 26.15538  |
| local_taxes   | -.956704    | .1379403                          | -6.94 | 0.000 | -1.227217            | -.6861907 |
| calendar_year |             |                                   |       |       |                      |           |
| 2014          | 2499028     | 6260933                           | 0.40  | 0.690 | -9779218             | 1.48e+07  |
| 2015          | 1848775     | 6324292                           | 0.29  | 0.770 | -1.06e+07            | 1.43e+07  |
| 2016          | 2712811     | 6574661                           | 0.41  | 0.680 | -1.02e+07            | 1.56e+07  |
| 2017          | -3697988    | 6952223                           | -0.53 | 0.595 | -1.73e+07            | 9935939   |
| 2018          | -4038495    | 6625908                           | -0.61 | 0.542 | -1.70e+07            | 8955500   |
| 2019          | -6561880    | 6987025                           | -0.94 | 0.348 | -2.03e+07            | 7140297   |
| 2020          | -7254732    | 6814128                           | -1.06 | 0.287 | -2.06e+07            | 6108379   |
| 2021          | 4151582     | 6258424                           | 0.66  | 0.507 | -8121744             | 1.64e+07  |
| 2022          | -2700332    | 6061453                           | -0.45 | 0.656 | -1.46e+07            | 9186717   |
| 2023          | 2601491     | 6217040                           | 0.42  | 0.676 | -9590677             | 1.48e+07  |
| _cons         | -1.87e+07   | 4673745                           | -4.01 | 0.000 | -2.79e+07            | -9569777  |
| sigma_u       | 52823354    |                                   |       |       |                      |           |
| sigma_e       | 65090374    |                                   |       |       |                      |           |
| rho           | .39708005   | (fraction of variance due to u_i) |       |       |                      |           |

# Clustering Standard Errors

|              | (1)<br>gen_gov        | (2)<br>gen_gov        | (3)<br>gen_gov        |
|--------------|-----------------------|-----------------------|-----------------------|
| fed_aid      | 17.92***<br>(36.23)   | 20.55***<br>(34.20)   | 20.55***<br>(6.28)    |
| state_aid    | 26.77***<br>(22.06)   | 23.39***<br>(16.59)   | 23.39**<br>(2.72)     |
| local_taxes  | -1.468***<br>(-12.48) | -0.957***<br>(-6.94)  | -0.957<br>(-1.43)     |
| 2013.calen~r |                       | 0<br>(.)              | 0<br>(.)              |
| 2014.calen~r |                       | 2499028.0<br>(0.40)   | 2499028.0<br>(1.06)   |
| 2015.calen~r |                       | 1848774.9<br>(0.29)   | 1848774.9<br>(0.79)   |
| 2016.calen~r |                       | 2712810.6<br>(0.41)   | 2712810.6<br>(1.08)   |
| 2017.calen~r |                       | -3697988.1<br>(-0.53) | -3697988.1<br>(-1.08) |
| 2018.calen~r |                       | -4038494.8<br>(-0.61) | -4038494.8<br>(-0.72) |

(1) Random Effects

(2) FE (no cluster)

(3) FE (cluster)

|                                  |                           |                           |                           |
|----------------------------------|---------------------------|---------------------------|---------------------------|
| 2017.calen~r                     | -3697988.1<br>(-0.53)     | -3697988.1<br>(-1.08)     |                           |
| 2018.calen~r                     | -4038494.8<br>(-0.61)     | -4038494.8<br>(-0.72)     |                           |
| 2019.calen~r                     | -6561879.7<br>(-0.94)     | -6561879.7<br>(-1.22)     |                           |
| 2020.calen~r                     | -7254731.9<br>(-1.06)     | -7254731.9<br>(-1.03)     |                           |
| 2021.calen~r                     | 4151582.2<br>(0.66)       | 4151582.2<br>(0.68)       |                           |
| 2022.calen~r                     | -2700332.4<br>(-0.45)     | -2700332.4<br>(-0.47)     |                           |
| 2023.calen~r                     | 2601490.8<br>(0.42)       | 2601490.8<br>(0.36)       |                           |
| _cons                            | -14813148.0***<br>(-8.45) | -18735405.9***<br>(-4.01) | -18735405.9***<br>(-3.38) |
| N                                | 2964                      | 2964                      | 2964                      |
| t statistics in parentheses      |                           |                           |                           |
| * p<0.05, ** p<0.01, *** p<0.001 |                           |                           |                           |

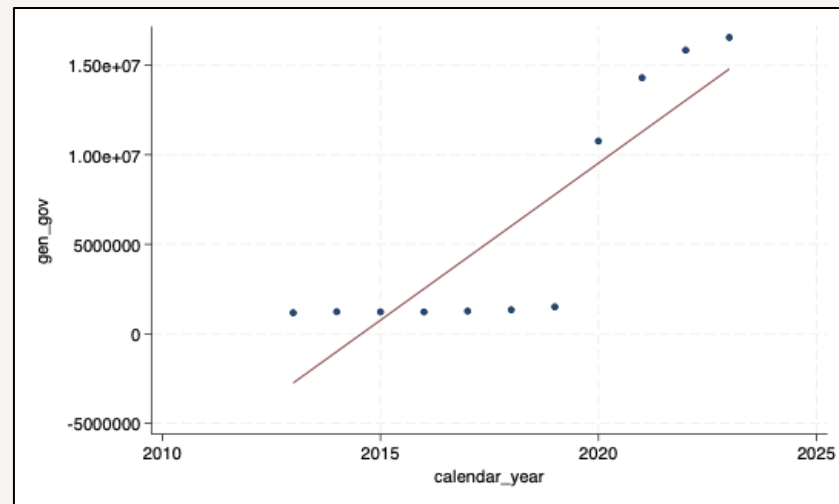
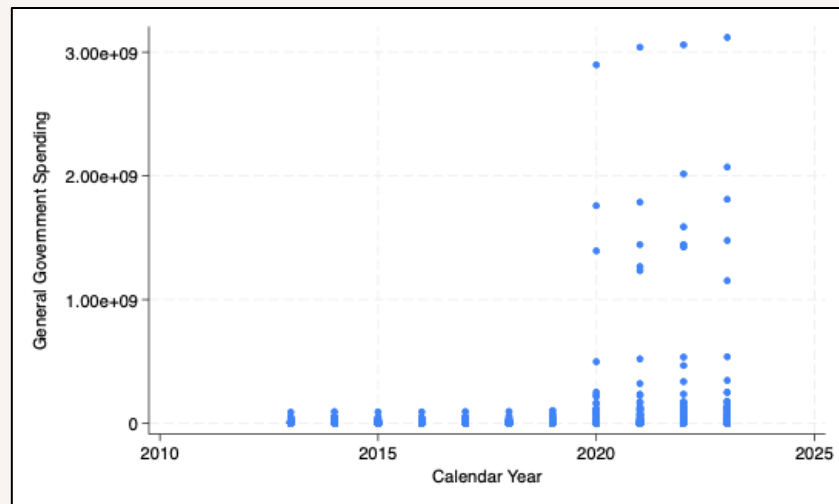
# Clustering Standard Errors

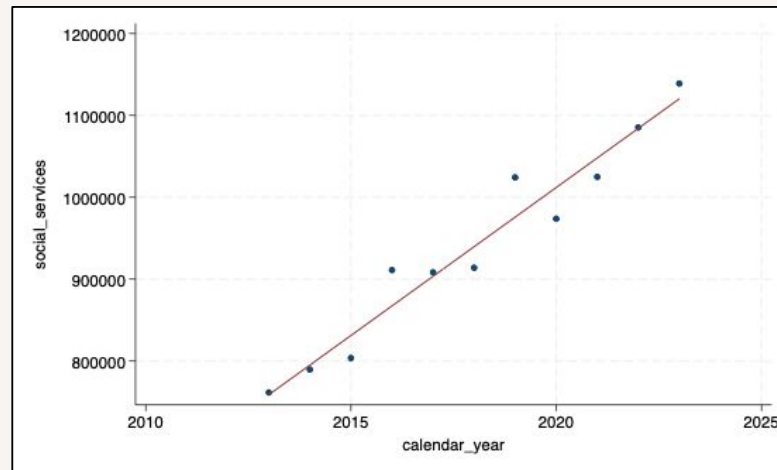
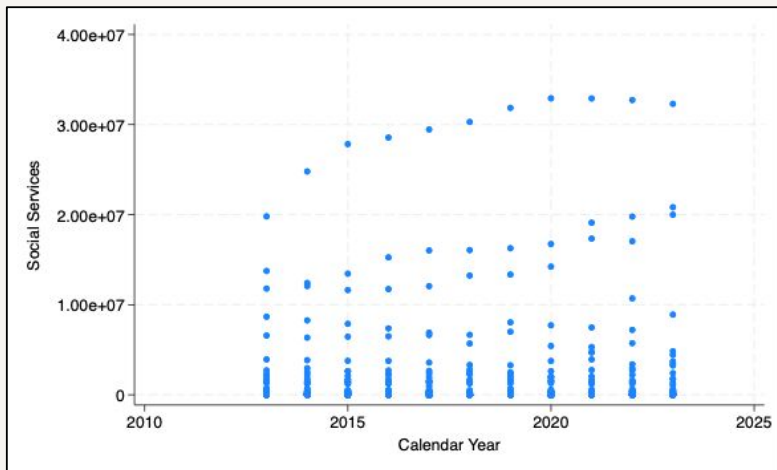
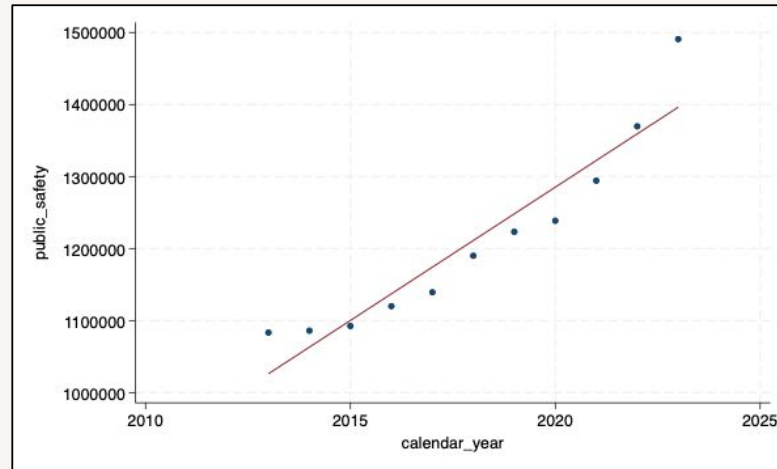
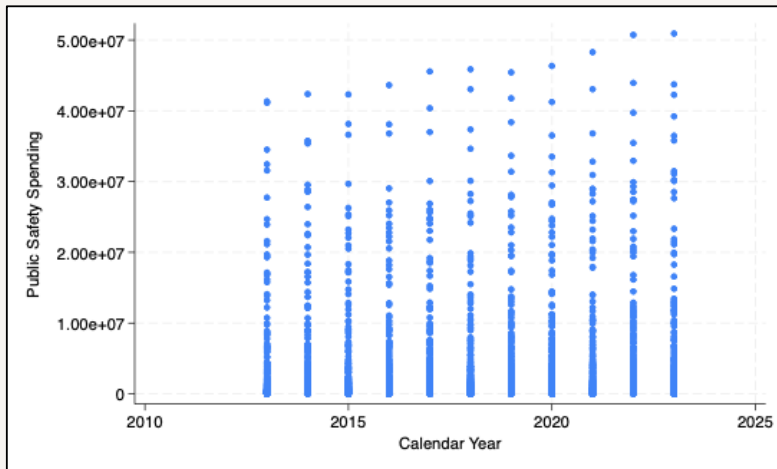
## Potential Unit Fixed Effects:

- Sense of community within town
- Urban vs Rural status
- Vacation town status

## Potential Time Fixed Effects:

- COVID
- Political Regime changes
- Spending Contributions to Pensions (Large spike in 2019)





# Findings

|              | (1)<br>gen_gov        | (2)<br>public_saf~y  | (3)<br>social_ser~s |
|--------------|-----------------------|----------------------|---------------------|
| fed_aid      | 20.55***<br>(6.28)    | -0.0759<br>(-1.73)   | 0.189**<br>(2.97)   |
| state_aid    | 23.39**<br>(2.72)     | -0.878***<br>(-5.60) | -0.199**<br>(-2.72) |
| local_taxes  | -0.957<br>(-1.43)     | 0.275***<br>(12.40)  | 0.0213***<br>(3.74) |
| 2013.calen~r | 0<br>(.)              | 0<br>(.)             | 0<br>(.)            |
| 2014.calen~r | 2499028.0<br>(1.06)   | -122505.3<br>(-1.43) | 119726.0<br>(0.94)  |
| 2015.calen~r | 1848774.9<br>(0.79)   | -135308.6<br>(-1.17) | 203952.6<br>(1.07)  |
| 2016.calen~r | 2712810.6<br>(1.08)   | -29084.7<br>(-0.27)  | 251482.5<br>(1.20)  |
| 2017.calen~r | -3697988.1<br>(-1.08) | 50542.5<br>(0.29)    | 299150.9<br>(1.15)  |
| 2018.calen~r | -4038494.8<br>(-0.72) | 8741.2<br>(0.04)     | 719509.2<br>(1.77)  |
| 2019.calen~r | -6561879.7<br>(-1.22) | 277918.1<br>(1.27)   | 830983.0<br>(1.72)  |

|                                  |                           |                      |                     |
|----------------------------------|---------------------------|----------------------|---------------------|
| 2016.calen~r                     | 2712810.6<br>(1.08)       | -29084.7<br>(-0.27)  | 251482.5<br>(1.20)  |
| 2017.calen~r                     | -3697988.1<br>(-1.08)     | 50542.5<br>(0.29)    | 299150.9<br>(1.15)  |
| 2018.calen~r                     | -4038494.8<br>(-0.72)     | 8741.2<br>(0.04)     | 719509.2<br>(1.77)  |
| 2019.calen~r                     | -6561879.7<br>(-1.22)     | 277918.1<br>(1.27)   | 830983.0<br>(1.72)  |
| 2020.calen~r                     | -7254731.9<br>(-1.03)     | 242602.1<br>(1.31)   | 796100.3<br>(1.40)  |
| 2021.calen~r                     | 4151582.2<br>(0.68)       | 635709.4**<br>(2.93) | 767891.2<br>(1.86)  |
| 2022.calen~r                     | -2700332.4<br>(-0.47)     | 445771.3*<br>(2.20)  | 1055117.9<br>(1.85) |
| 2023.calen~r                     | 2601490.8<br>(0.36)       | 457935.5*<br>(2.33)  | 810872.6<br>(1.78)  |
| _cons                            | -18735405.9***<br>(-3.38) | 337351.6*<br>(2.38)  | 10466.0<br>(0.03)   |
| N                                | 2964                      | 2964                 | 604                 |
| t statistics in parentheses      |                           |                      |                     |
| * p<0.05, ** p<0.01, *** p<0.001 |                           |                      |                     |



# Findings

|              | (1)<br>gen_gov        | (2)<br>public_saf~y  | (3)<br>social_ser~s |
|--------------|-----------------------|----------------------|---------------------|
| fed_aid      | 20.55***<br>(6.28)    | -0.0759<br>(-1.73)   | 0.189**<br>(2.97)   |
| state_aid    | 23.39**<br>(2.72)     | -0.878***<br>(-5.60) | -0.199**<br>(-2.72) |
| local_taxes  | -0.957<br>(-1.43)     | 0.275***<br>(12.40)  | 0.0213***<br>(3.74) |
| 2013.calen~r | 0<br>(.)              | 0<br>(.)             | 0<br>(.)            |
| 2014.calen~r | 2499028.0<br>(1.06)   | -122505.3<br>(-1.43) | 119726.0<br>(0.94)  |
| 2015.calen~r | 1848774.9<br>(0.79)   | -135308.6<br>(-1.17) | 203952.6<br>(1.07)  |
| 2016.calen~r | 2712810.6<br>(1.08)   | -29084.7<br>(-0.27)  | 251482.5<br>(1.20)  |
| 2017.calen~r | -3697988.1<br>(-1.08) | 50542.5<br>(0.29)    | 299150.9<br>(1.15)  |
| 2018.calen~r | -4038494.8<br>(-0.72) | 8741.2<br>(0.04)     | 719509.2<br>(1.77)  |
| 2019.calen~r | -6561879.7<br>(-1.22) | 277918.1<br>(1.27)   | 830983.0<br>(1.72)  |

General Government Spending  
mainly driven by state and fed aid

Public Safety mainly driven by  
Local taxes

Social Services mainly fed aid, but  
a little bit of local taxes.

# Findings

Within towns in NY:

- Most of the general spending comes from government aid
- Local taxes are the main contributor to public safety spending
- Federal aid and local taxes contribute highest to social services spending

Extensions:

- Expand the data outside of just NY, possibly Tri-state area or entire US
- Expand data for more years than just from 2013-2023
- Control for more variables (population, political regime, median income)

Thank you